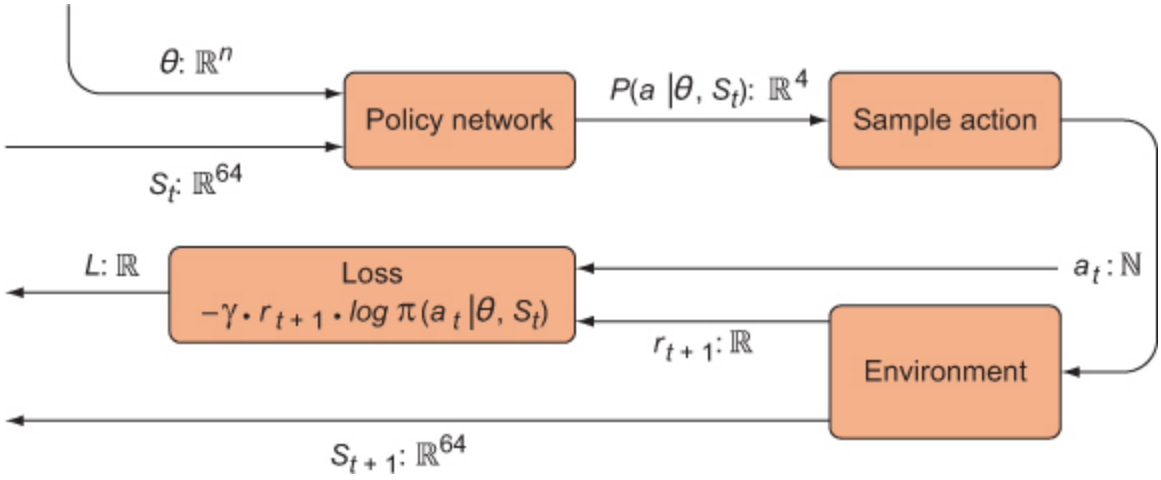
1)$\log\pi(a_1|\theta,S_0) = \log\pi(a_1|\theta,S_0)$, which is the log-probability output from the policy network (see figure 4.7).

Figure 4.7. A string diagram for training a policy network for Gridworld. The policy network is a neural network parameterized by θ (the weights) that accepts a 64-dimensional vector for an input state. It produces a discrete 4-dimensional probability distribution over the actions. The sample action box samples an action from the distribution and produces an integer as the action, which is given to the environment (to produce a new state and reward) and to the loss function so we can reinforce that action. The reward signal is also fed into the loss function, which we attempt to minimize with respect to the policy network parameters.



4.3. Working with OpenAI Gym

To illustrate how policy gradients work, we've been using Gridworld as an example, since it is already familiar to you from last chapter. However, we should use a different problem to actually implement the policy gradient algorithm, both for variety and also to introduce the OpenAI Gym.

The OpenAI Gym is an open source suite of environments with a common API that is perfect for testing reinforcement learning algorithms. If you come up with some new DRL algorithm, testing it on a few of the environments in the Gym is a great way to get some idea of how well it performs. The Gym contains a variety of environments from easy ones can be "solved" by simple linear regression all the way through to ones that all but require a sophisticated DRL approach (see figure 4.8). There are games,